

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0532

Duck Creek 138 kV
November 21, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0532. The Interconnection Customer proposes a 413.7 MW natural gas combined cycle facility interconnecting at the Duck Creek 138 kV substation in the Pennsylvania area. The request was received on October 29, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 187 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$43,539,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DUCK CREEK / BAYFIELD 138 kV L	MACON SOLAR POWER PROJECT / DUCK CREEK 138 kV Line	158.9	191	64.4
PINEY / BAYFIELD 138 kV	MACON SOLAR POWER PROJECT / DUCK CREEK 138 kV	133.2	221	64.4
CICO / DUCK CREEK 138 kV Line	DUCK CRK / BAYFIELD 138 kV	121.5	191	63.2
PINEY / PINEY CREEK PROJECT 138 kV Line	DUCK RK / MACON SOLAR POWER PROJECT 138 kV	105.6	240	56.9
BLUEBONNET / RUSSELL FORK 230 kV L	DEKALB / JONES CHAPEL 500 kV	100.2	900	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DUCK CREEK / BAYFIELD 138 kV L	Rebuild line with higher-capacity conductor	\$2,479,000	95% (\$2,344,000)
BAYFIELD / PINEY 138 kV	Rebuild line with higher-capacity conductor	\$3,867,000	95% (\$3,657,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CICO / DUCK CREEK 138 kV Line	Rebuild line with higher-capacity conductor	\$35,741,000	97% (\$34,517,000)
PINEY / PINEY CREEK PROJECT 138 kV	Reconductor line	\$1,034,000	94% (\$968,000)
BLUEBONNET / RUSSELL FORK 230 kV	Reconductor line	\$19,321,000	11% (\$2,053,000)
	Total allocated to project		\$43,539,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.